Why quantum computing?



Day 0 · Saturday, Oct 10

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Nature isn't
classical,
dammit

The problem: simulating nature



“Nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical.”
Richard Feynman, 1981

Molecules, materials and spin models are quantum systems. Describing n interacting particles classically takes resources that grow exponentially with n.
A quantum computer is a controllable quantum system, so it can mimic them directly. This is the problem we are answering.

n qubits are described by 2^n complex amplitudes. n classical bits need just n numbers.

2^n

Each extra qubit doubles the state space a classical computer must track.

$\times 2$

How many qubits can classical machines simulate?

30

qubits fit in a modern laptop

48

qubits fit in the IBM Summit supercomputer

60

qubits exceed all classical computers on Earth connected

Source: IBM QGSS 2026, Lecture 1. IBM Starling (2029) targets 200 logical qubits.

Speedups are not free parallelism

1

Superposition. A state combines many basis states, but a measurement returns only one outcome.

2

Entanglement. Information is shared jointly across qubits instead of being stored in each one.

3

Interference. Amplitudes combine so wrong answers cancel and right answers reinforce.

4

Hidden structure. Algorithms exploit global structure of a problem, such as symmetry or correlations.

Three problem areas, one week

Simulating nature

Spin models, materials, chemistry, high-energy physics.

Day 2: VQE.

Math and structured data

Factoring, hidden subgroups, linear algebra, learning.

Day 4: QFT and phase estimation.

Search and optimization

Combinatorial and black-box optimization.

Day 3: IBM industry expert talk.

Secure communication

Quantum states can also protect information.

Day 4: quantum key distribution.



Hardware noise: about 0.1% error per gate, so deeper circuits fail more often.

0.1%

Utility era since 2023: 100+ qubits and over 5,000 two-qubit gates.

>5k

Source: IBM QGSS 2026, Lecture 1

Utility is here, advantage is next

Quantum utility (2023)

A quantum computer runs circuits beyond the ability of a classical computer to simulate exactly.

Quantum advantage (not yet)

A quantum computer solves a problem more accurately, cheaper or more efficiently than classical computing alone.



200 logical qubits by 2029



The rest of the week

Day 1

Quantum foundations, Qiskit 101 and hands-on labs.

Day 2

VQE and Qiskit MCP.

Day 3

IBM industry expert talk and faculty panel.

Day 4

Quantum key distribution, QFT and phase estimation.

Now, set up your environment

